

$$I = \int_0^\pi e^{x \sin n} dx \quad \forall n \in \mathbb{N}, e^n = \sum_{k=0}^{\infty} \frac{n^k}{k!}$$

$$\forall n \in \mathbb{N}, e^{x \sin n} = \sum_{k=0}^{\infty} \frac{(x \sin n)^k}{k!}$$

$$\begin{aligned} \forall n \in \mathbb{N} \quad I &= \int_0^\pi \sum_{k=0}^{\infty} \frac{(x \sin n)^k}{k!} dx \\ &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} \int_0^\pi (x \sin n)^k dx \right) + \int_0^\pi \frac{x \sin n}{1!} dx + \int_0^\pi \frac{x^2 \sin^2 n}{2!} dx \\ &= \sum_{k=2}^{\infty} \frac{1}{k!} J_k + \pi + 2 \end{aligned}$$

$$\begin{aligned} \forall k \in \mathbb{N}^* \quad \forall n \in \mathbb{N}, J_k &= - \int_0^\pi \cos n x \sin^{k-1} n x dx \\ &= - [\cos n x \sin^{k-1} n x]_0^\pi + \int_0^\pi (\cos n x) (k-1) \cos n x \sin^{k-2} n x dx \\ &= (k-1) \int_0^\pi (1 - \sin^2 n x) \sin^{k-2} n x dx \\ &= (k-1) (J_{k-2} - J_k) \end{aligned}$$

$$\text{Done } J_k = \frac{(k-1)}{k} J_{k-2}$$

$$\text{So } k = 2k' \quad (k' \in \mathbb{N}^*), J_{2k'} = \frac{2k'-1}{2k'} J_{2k'-2} = \frac{(2k'-1) \times \dots \times 1}{(2k') \times \dots \times 2} J_{2k'-2k'}$$

$$= \frac{(2k'-1)(2k'-3) \times \dots \times 3 \times 1}{(2k')(2k'-2) \times \dots \times 4 \times 2} J_{2k'-2k'}$$

$$= \frac{(2k'-1)!}{2k'!} J_0$$

$$= \frac{\pi (2k'-1)!}{2k'! \times (2^{k'-1} \times (k'-1)! \times \dots \times 1)^2}$$

$$\text{Done } J_{2k'} = \frac{\pi (2k'-1)!}{k'! 2^{k'-1} [(k'-1)!]^2}$$

$$\text{So } k = 2k' + 1 \quad (k' \in \mathbb{N}^*), J_{2k'+1} = \frac{2k'}{2k'+1} J_{2k'-1}$$

$$= \frac{(2k')(2k'-2) \times \dots \times 4 \times 2}{(2k'+1)(2k'-1) \times \dots \times 5 \times 3} J_{2k'+1-1}$$

$$= \frac{(2k')(2k'-2) \times \dots \times 4 \times 2}{(2k'+1)!} \times J_1$$

$$= \frac{(2k'! \times k'! \times (k'-1)! \times \dots \times 2 \times 1)^2}{(2k'+1)!} \times 2$$

$$= \frac{2^{2k'+1} (k'!)^2}{(2k'+1)!}$$

$$(2k'+1)!$$

$$I = \int_0^\pi e^{x \sin n} dx = ?$$

$$\text{aussi } I = \sum_{k=2}^{+\infty} \frac{1}{k!} J_k + \pi + 2$$

$$= \pi + 2 + \sum_{i=1}^{+\infty} \frac{1}{(2i)!} J_{2i} + \sum_{j=1}^{+\infty} \frac{1}{(2j+1)!} J_{2j+1}$$

$$= \pi + 2 + \sum_{i=1}^{+\infty} \frac{1}{(2i)!} \times \frac{\pi(2i-1)!}{i \times 2^{2i-1} \times [(i-1)!]^2} + \sum_{j=1}^{+\infty} \frac{1}{(2j+1)!} \times \frac{2^{2j+1} \times (j!)^2}{(2j+1)!}$$

$$= \pi + 2 + 2\pi \sum_{i=1}^{+\infty} \frac{1}{(2i) \times i \times 2^{2i} \times (i-1) \times (i-1)!} + 2 \sum_{j=1}^{+\infty} \frac{2^{2j} (j!)^2}{[(2j+1)!]^2}$$

$$\text{Donc } I = \int_0^\pi e^{\cos x} dx = \pi \left(1 + \sum_{i=1}^{+\infty} \frac{1}{2^{2i} (i!)^2} \right) + 2 \left[1 + \sum_{j=1}^{+\infty} 2^{2j} \left[\frac{j!}{(2j+1)!} \right]^2 \right]$$

ou encore

$$I = \int_0^\pi e^{\cos x} dx = \pi \sum_{i=0}^{+\infty} \frac{1}{2^{2i} (i!)^2} + 2 \sum_{j=0}^{+\infty} 2^{2j} \left[\frac{j!}{(2j+1)!} \right]^2$$

ou encore

$$I = \int_0^\pi e^{\cos x} dx = \sum_{k=0}^{+\infty} \left[\frac{\pi}{2^{2k} (k!)^2} + 2^{2k+1} \left[\frac{k!}{(2k+1)!} \right]^2 \right] = \sum_{k=0}^{+\infty} \frac{\pi}{(2^k k!)^2} + 2 \times \left(\frac{2^k k!}{(2k+1)!} \right)^2$$

~~$m = e$~~

$$I_k = \int_a^b (m \ln m)^k dm$$

$$= \frac{1}{2} \left[\frac{e^{2b}}{2} - \frac{e^{2a}}{2} \right] = \frac{1}{2} \left[\frac{e^{2b}}{2} - \frac{e^{2a}}{2} \right]$$

$$= \frac{1}{k+1} \left[\frac{m^{k+1}}{k+1} \right]_a^b = \frac{1}{k+1} \left[\frac{b^{k+1}}{k+1} - \frac{a^{k+1}}{k+1} \right]$$

$$= \frac{1}{k+1} \left[A - K \int_a^b m^k \ln m dm \right]$$

On pourrait poser $J_k(m) = \int_a^m x^k \ln x dx$ $m \in \mathbb{R}$ $k \in \mathbb{N}$

et là on aurait $J_0(m) = m$ $J_1(m) = 1 - \cos m$

et $\forall k \in \mathbb{N}^* (1,1)$, $J_k(m) = \frac{(k-1) J_{k-2} + \cos m \cdot m^{k-1}}{k}$

et utiliser alors $J_k(m)$.

En écrivant $m = \frac{e^{i\theta} + e^{-i\theta}}{2}$ on pourrait encore remarquer que

$$I = \sum_{k=0}^{+\infty} \frac{1}{k!} J_k = \sum_{k=0}^{+\infty} \frac{1}{k!} \frac{1}{2^{k+1}} \sum_{j=0}^k \frac{C_k^j (-1)^j (e^{2j-k} - 1)}{(2j-k)!} \text{ et procéder au calcul terme par terme ou } 2j-k=0$$